

Introduction to Macroeconomics (S102016CR)

Spring Semester

<https://moodle.unige.ch/course/view.php?id=6618>

Marcelo Olarreaga, Office M5270

Introduction to Macroeconomics

Part I
National Accounts and
the Real Economy
(Weeks 1 to 3)

Part II
Monetary Economics and
Financial Economics
(Weeks 4 to 6)

Part III
Open Macroeconomics
(Weeks 7 to 9)

Part IV
Macroeconomic Policies
(Weeks 10 to 12)

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graph TD; A["First Part:  
National Accounts  
and the Real  
Economy"] --> B["Measuring a  
nation's income  
(Chapter 1)"]; A --> C["Measuring the cost  
of living  
(Chapter 2)"]; A --> D["Unemployment  
(Chapter 3)"]
```

First Part:
National Accounts
and the Real
Economy

Measuring a
nation's income
(Chapter 1)

Measuring the cost
of living
(Chapter 2)

Unemployment
(Chapter 3)

First Part:
National Accounts
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**Measuring a
nation's income
(Chapter 1)**

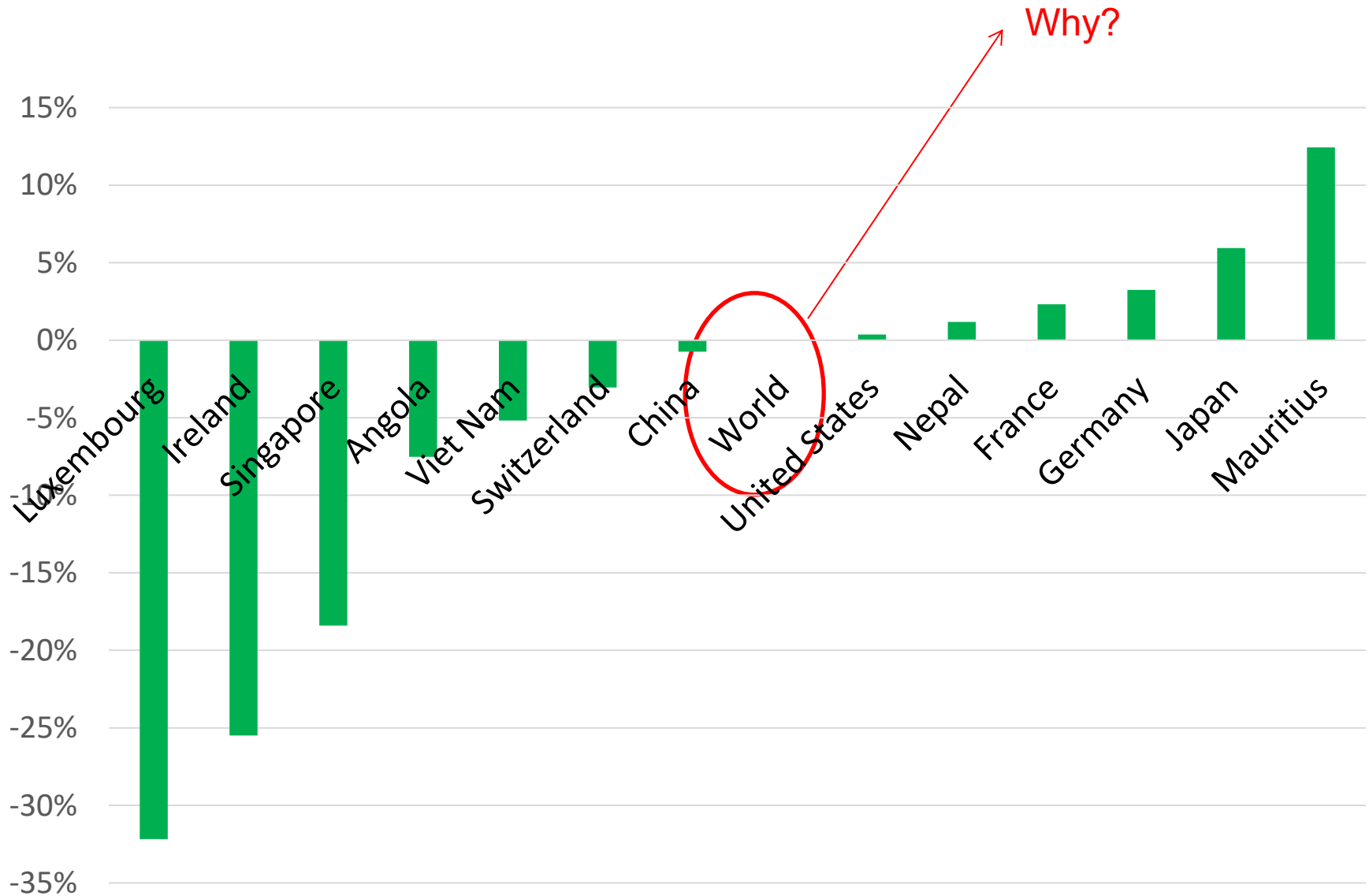
**Measuring the cost
of living
(Chapter 2)**

Unemployment
(Chapter 3)

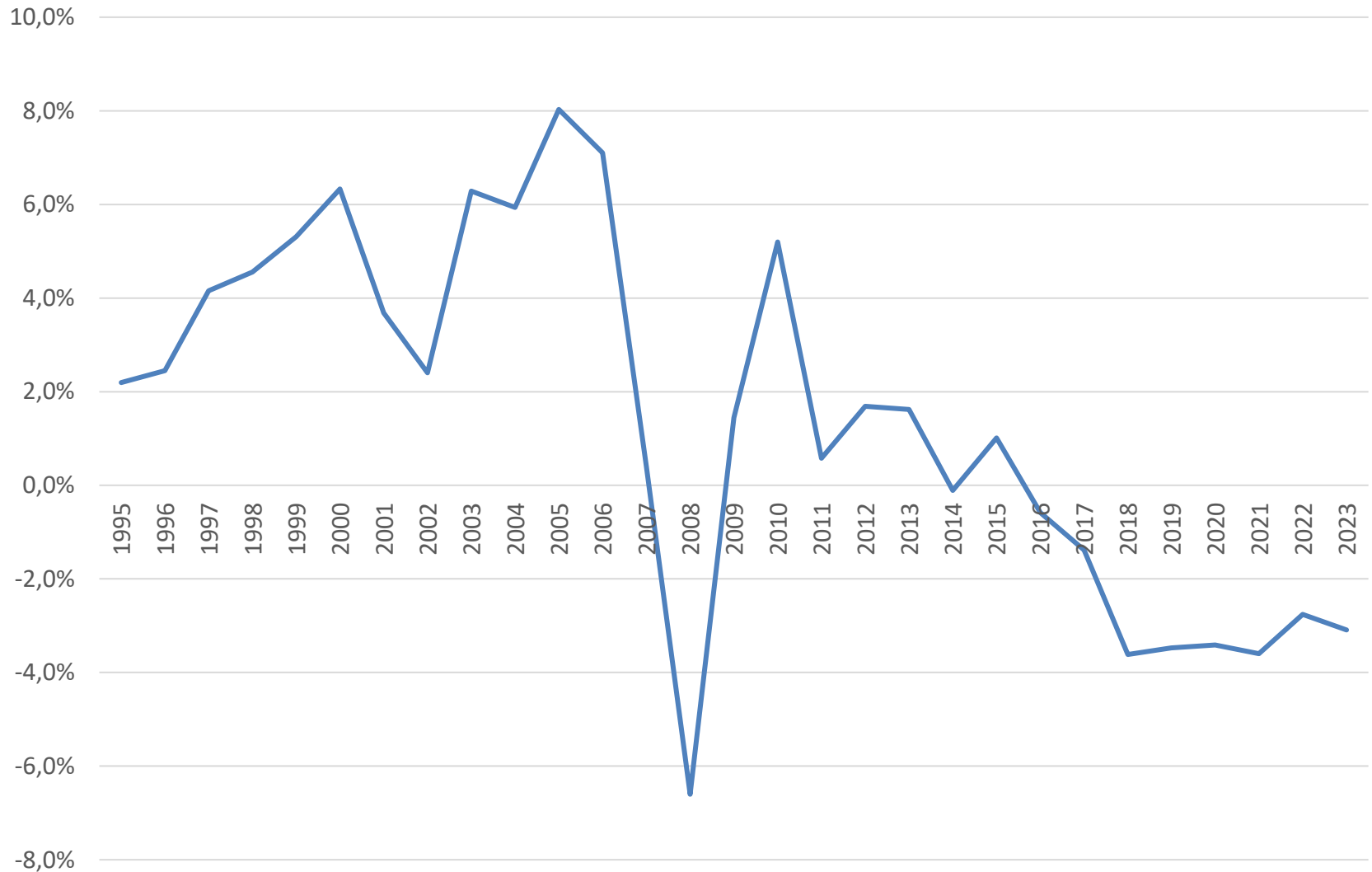
Friday session:
For Weeks 1 and 2

1. Videos for all classes are available online.
 - Tuesdays classes are a set of 4 to 5 videos per week already available on Moodle.
 - Tuesdays Q&A sessions are available on Mediaserver after each class (No new material)
 - Friday classes are available on Mediaserver after each class
2. Please use the FAQ for each chapter on Moodle to ask questions on class content

« GNP/GDP ratios – 1 » in 2023 across different countries



Ratio of GNP to GDP in Switzerland ($\text{GNP}/\text{GDP} - 1 = \text{NFIA}/\text{GDP}$)



What happened in 2009? And in 2011?

From GDP, to GNP to NNP to NDP to National Income...etc...

The production approach to measure GDP as the sum of each firm value-added

Final Good (beer)

Production = 100

Intermediate good (wheat)

Production = 50

GDP?

The production approach to measure GDP as the sum of each firm value-added

Final Good (beer)

Production = 100

Intermediate purchases = 50

Value-added = production – intermediate purchases
 $= 100 - 50 = 50$

Intermediate good (wheat)

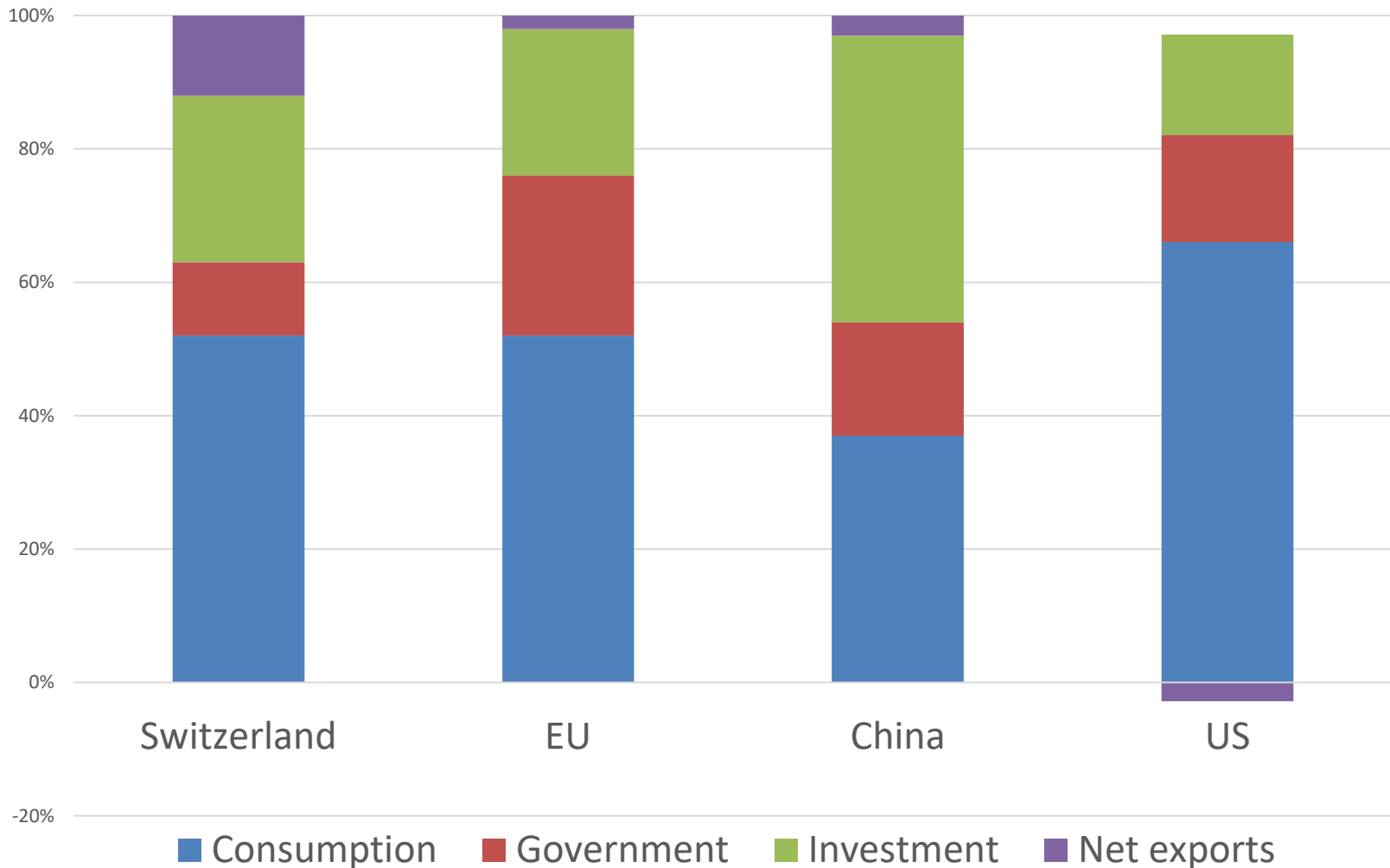
Production = 50

Intermediate purchases = 0

Value added = production – intermediate purchases
 $= 50 - 0 = 50$

$GDP = \text{Value added beer} + \text{Value added wheat} = 50 + 50 = 100$

Components of Aggregate Demand 2024

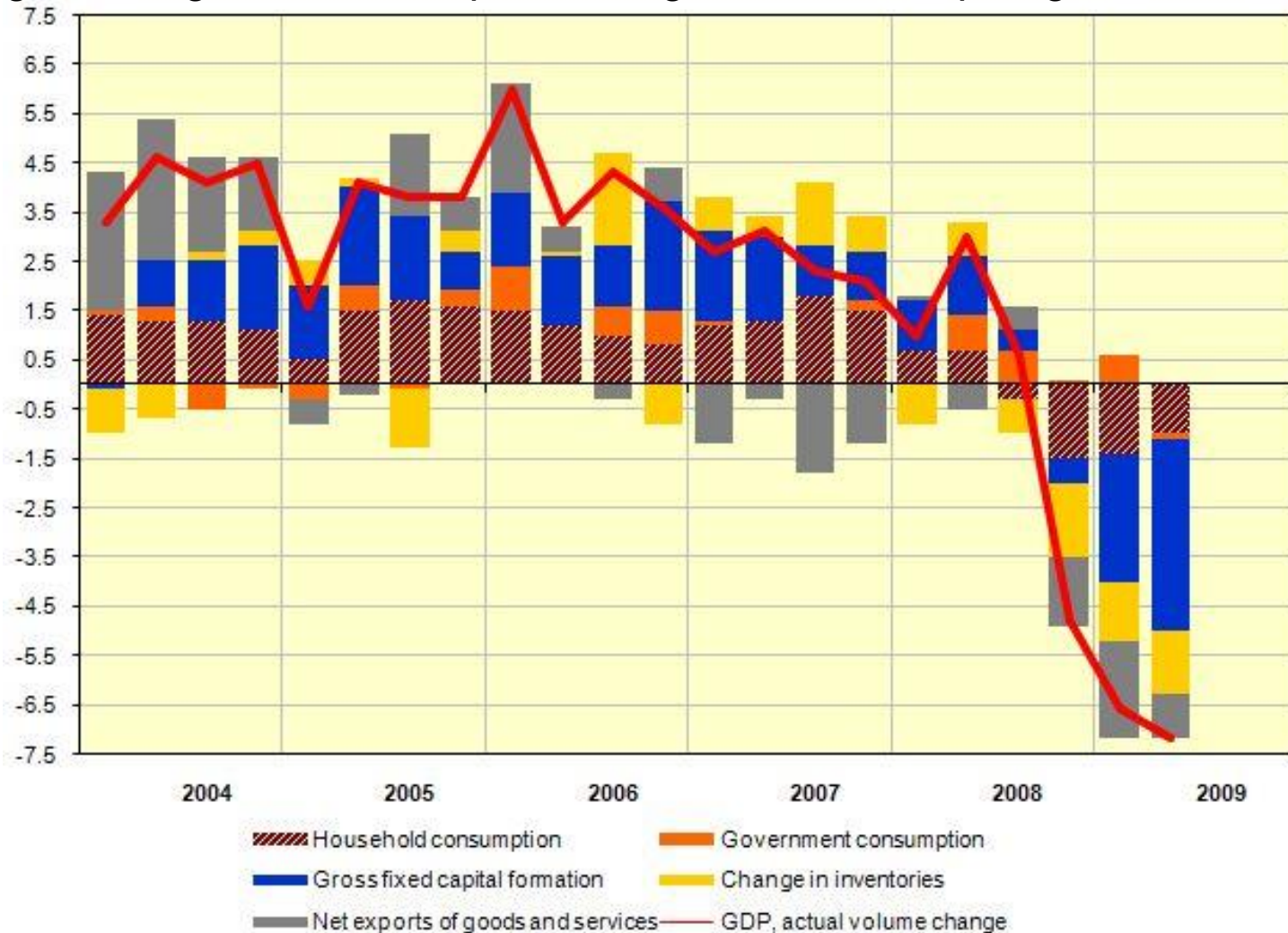


Source: World Bank's World Development Indicators, 2026

Contribution to GDP growth in the US (2004-2009, quarterly):

$$Y\% = C/Y*(C\%) + I/Y*(I\%) + G/Y*(G\%) + (X-M)/Y*([X-M]\%)$$

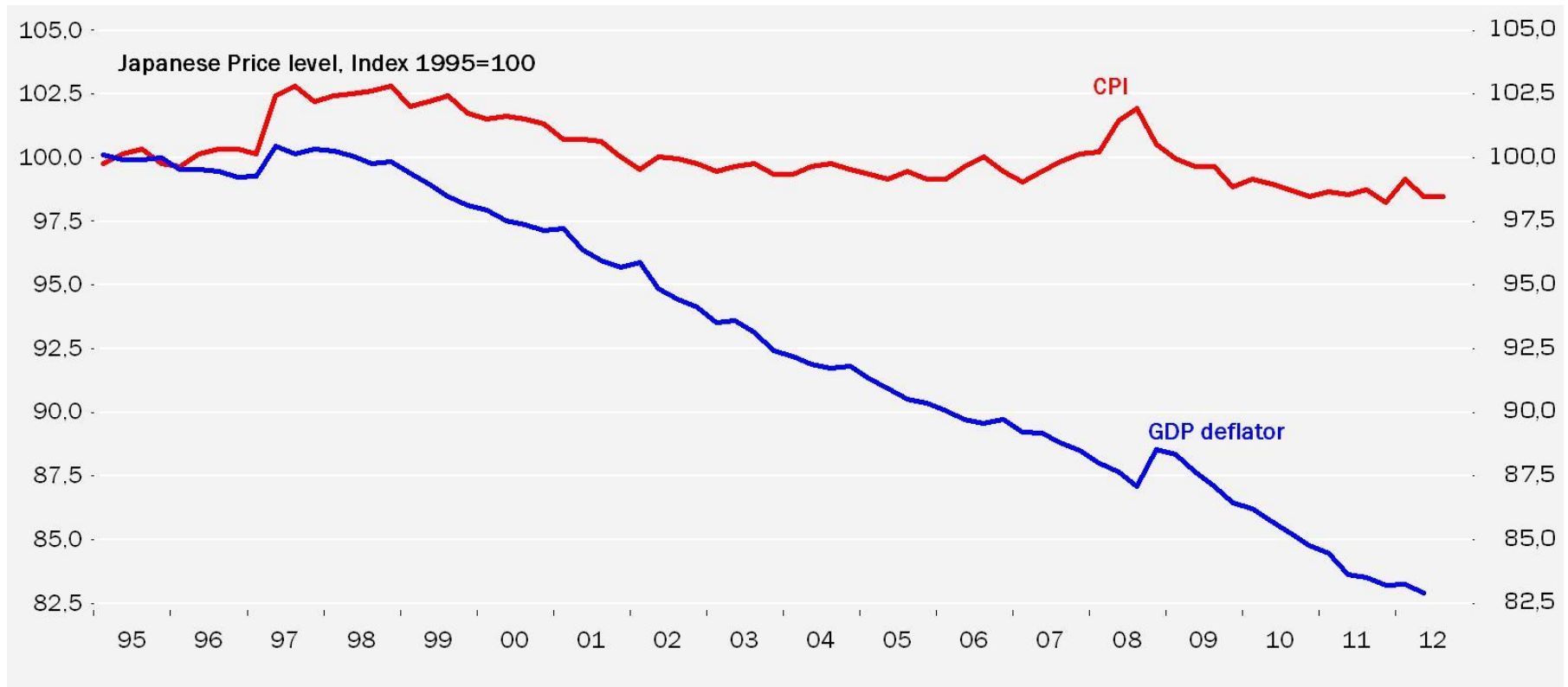
GDP growth = contribution of consumption growth + investment growth + government expenditure growth + net export growth



In 2015 Ireland's (nominal) GDP increased by 32%, GNI by 9%, NNI by 6%....

- Read OECD (2016) in the Additional Documents Folder on Moodle
- Clearly, Ireland was doing well in 2015
- But there is a difference between 32 and 6%
- Can you explain these differences?
- What are the implications for policymakers?
- What is the difference between GNP and GNI?

CPI and GDP deflator evolution in Japan



What drove this difference? Shall BOJ worry about inflation measured by CPI or GDP deflator?

Formally explaining the (ex-ante) Fisher parity:

$$r = i - \pi$$

- The nominal return of 1 CHF measured in monetary units is $(1 + i)$ where i is the nominal interest rate
- The real return of 1 CHF measured in monetary units is $(1 + r)$ where r is the real nominal interest rates.
- Then as usual:

$$\text{Nominal return}_t = \text{Real return}_t P_t/P_0 = \text{Real return}_t * (1+\text{inflation})_{t+1}$$

or

$$\text{Real return}_t = \text{Nominal return}_t / (1+\text{inflation})_{t+1}$$

- We can rewrite this as :

$$\text{Value} (1+r_t) = \text{Value} (1+i_t) / (1+\text{inflation}_{t+1})$$

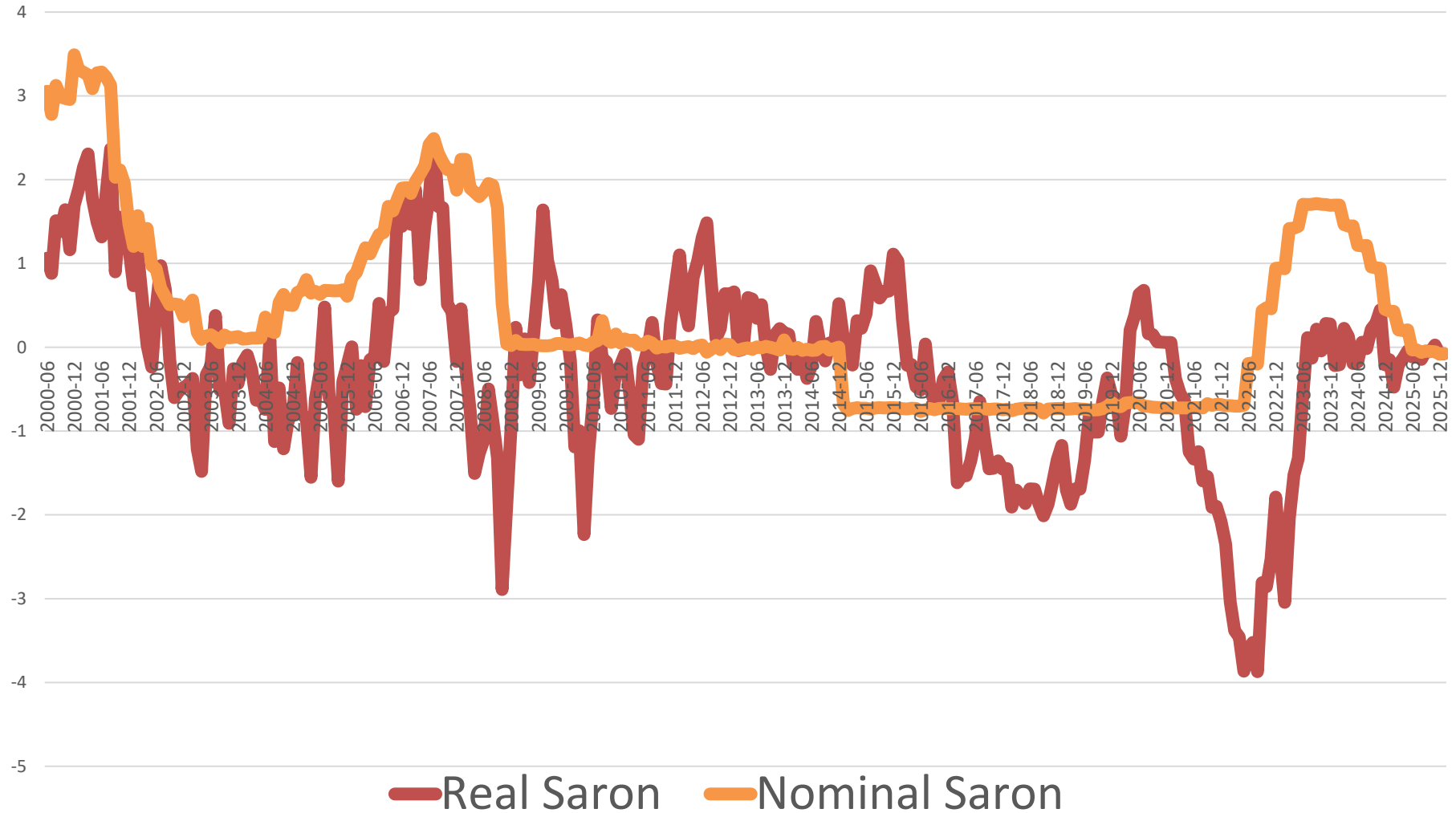
$$(1+r_t) = (1+i_t) / (1+\pi_{t+1})$$

- Note that π_{t+1} is not observed at t . It has to be anticipated/expected.
- Taking logs on both sides we have:

$$\ln (1+r_t) = \ln (1+i_t) - \ln (1+\pi_{t+1})$$

- Recalling that for small x , the $\ln (1+x) \approx x$ (a linear approximation, as by the definition of a derivative: $f'(0) = (f(x)-f(0))/(x-0)$, which implies $f(x) \approx f(0) + f'(0)(x-0)$, so $\ln(1+x) = \ln (1 + 0) + 1/(1+0) (x-0) = x$)
- We then have the ex-ante Fischer parity: $r_t \approx i_t - \pi_{t+1}$
- When computed after inflation has been observed we talked of ex-post Fischer parity: $r_{t-1} \approx i_{t-1} - \pi_t$

When are real interest rates negative? And higher than nominal rates?
Can nominal interest rates be negative?



The CPI in year t is with base in year 0 given by:

$$\text{CPI}_{t,0} = \frac{\text{Cost in t of fixed bundle in 0}}{\text{Cost in 0 of fixed bundle in 0}} = \frac{\sum_i p_i^t c_i^0}{\sum_i p_i^0 c_i^0}$$

The problem is that if year t and year 0 are very far apart, the consumption bundle in year 0 may not reflect very well what is consumed in year t, and therefore is not a very good representation of changes in the cost of living.

The solution is to use more frequent updates of the consumption bundle to capture adjustment to price changes and introduction of new goods, and compute sub-indices (or link indices) for each previous period.

The result is a chained-CPI index that can be expressed as:

$$\text{Chained CPI}_{t,0} = \text{CPI}_{t,t-1} * \text{CPI}_{t-1,t-2} * \text{CPI}_{t-2,t-3} \dots \text{CPI}_{t-n,t-n+1}$$

The chained CPI partly corrects for the overestimation of the true increase in the cost of living due to substitution bias in the CPI, but it may underestimate the true cost of living because it allows full substitution in each period.

The Exam

Material covered

- All that has been seen in class or seminar
- The companion chapters in M&T or K&W if the material was covered in class or seminar
- No surprises in terms of material covered
- No differences between the English and French exams (apart from language).

Type of exam

- 2 hours
- 30 MCQ with 4 possible answers. No penalties → Answer all questions
- There will be 6 to 8 questions on the first part of the course, 6 to 8 questions on the second part, 6 to 8 questions on the third part, and 6 to 8 questions on the fourth part.
- Over the entire exam
 - 1/3 of the questions will require some simple calculus
 - 2/3 of the questions will be analytical
- A two-sided A4 page of personal notes and a non-programmable calculator are allowed

Examples of exam questions for weeks 1 and 2

votamatic.unige.ch

code d'accès: GTLN



<https://votamatic.unige.ch/form/C588KmjHAigZCkiktYMP>